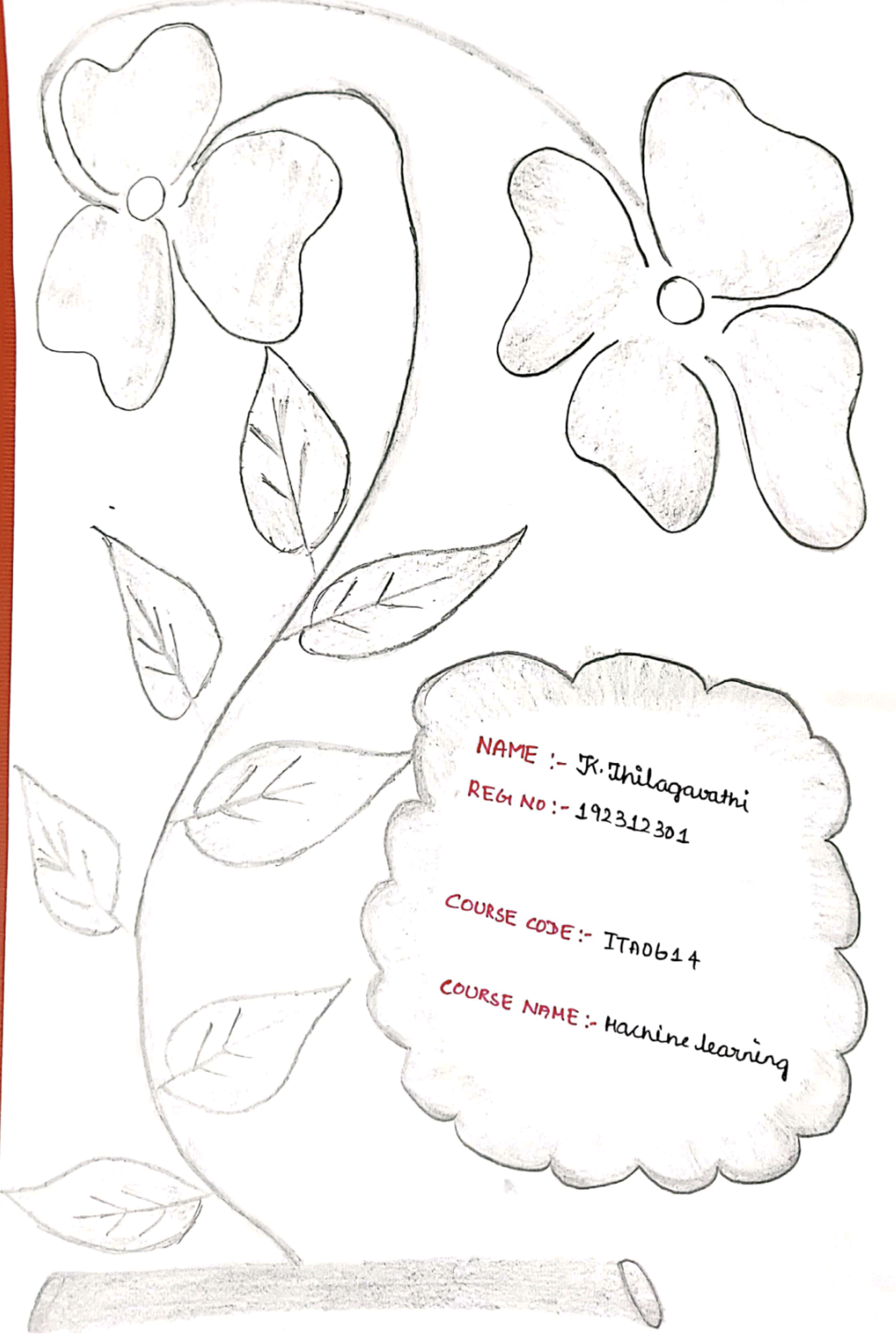

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COURSE NAME :- Machine learning

1. In a real-world medical image classification task for detecting breast cancer, a deep learning model integrated with the K-nearest neighbors algorithm is applied for binary classification (malignant Vs. begnign). the dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210 true positive (TP), 190 true Negatives (TN), and 30 false positives (FP). Based on the given information, determine the missing false Negatives (FN) value, and then calculate the Accuracy, precision, sensitivity (Recall), and specificity of the classifier, explaining its effectiveness in medical Diagnosis.

Given:-

TP = 210 FP = 30

TN = 190 FN = ?

Total data set = 1800 (malignant) + 1800 (benign)

= 3600 samples.

Test set (20%) = $\frac{20}{100} \times 3600 \Rightarrow 720$ samples.

False Negatives (FN)

FN = Total test − (TP + TN + FP)

FN = 720 − (210 + 190 + 30)

FN = 720 − 430

FN = 290

Accuracy

$$\text{Accuracy} = \frac{TP + TN}{T_{total}}$$

$$\text{Accuracy} = \frac{210+190}{720}$$

$$= \frac{400}{720}$$

$$\text{Accuracy} = 0.556\ (55.6\%)$$

Precision

$$\text{Precision} = \frac{TP}{TP+FP}$$

$$\text{Precision} = \frac{210}{210+30} = \frac{210}{240}$$

$$\text{Precision} = 0.875\ (87.5\%)$$

Sensitivity (Recall)

$$\text{Recall} = \frac{TP}{TP+FN}$$

$$\text{Recall} = \frac{210}{210+290} = \frac{210}{500}$$

$$\text{Recall} = 0.42\ (42\%)$$

Specificity:-

$$\text{Specificity} = \frac{TN}{TN+FP}$$

$$\text{Specificity} = \frac{190}{190+30} = \frac{190}{220} = 0.864\ (86.4\%)$$

In medical applications, high recall is more important than precision, because missing a cancer case is far more serious than a false alarm.

2) Examine the performance of a machine-learning model using its training and validation loss values over the multiple epochs :-

Epoch	Training loss	Validation loss
1	2.8	2.4
2	2.5	2.2
3	2.2	2.0
4	2.0	2.1
5	1.8	2.3
6	1.6	2.6
7	1.5	2.9
8	1.3	3.2

i) Identify the epoch at which the model begins to overfit.

⇒ up to Epoch 3 :

Training ↓ and validation ↓ → good learning.

⇒ From Epoch 4 onwards :

* Training loss ↓ (model Improving on the training data)
* validation loss ↑ (Performance worsening on unseen data
* overfitting starts at Epoch 4.

ii) Suggest two effective techniques to reduce overfitting in this model.

1. Early stopping.

* Stop training when validation loss starts increasing.
* In this case → stop at Epoch 3.

2. Regularization

* Add constraints to prevent model from the memorizing data.

Examples:-

* L_2 Regularization (Ridge)
* Dropout (in deep learning)

iii) Discuss how overfitting affects the model's generalization ability on unseen data.

* The model memorizes training data instead of learning pattern.
* performs: very well on training data and poorly on new/unseen data.
* In this case training loss keeps decreasing (1.3 at EPOCH 8) and validation loss increases (3.2 at EPOCH 8)
* This means model is not generalizing predictions on real-world data will be inaccurate.